



# Soknarith (Soki) Sem

☎ (401) 952-6819 — ✉ [seminaia401@gmail.com](mailto:seminaia401@gmail.com) — 🌐 [Seminaia](#) — in [soknarith-sem](#) — 📍 Worcester, MA

## SUMMARY

Systems modeling and simulation engineer with experience in MATLAB-based modeling, numerical methods, and physics-based simulation. Skilled in translating physical systems into computational models for analysis and prediction.

## WORK EXPERIENCE

**Research Assistant / Graduate Teaching Assistant** May 2024 to Present  
*Worcester Polytechnic Institute – Worcester, MA*

- Developed physics-based computational models using DFT and molecular dynamics to analyze system behavior and transport mechanisms.
- Applied numerical methods and data analysis to evaluate system response across varying operating conditions.
- Utilized MATLAB and Python for simulation post-processing, data analysis, and visualization of complex system behavior.
- Translated physical phenomena into computational frameworks to support predictive modeling and system-level understanding.

**Chemical Process Engineer** May 2023 to May 2024  
*Knowles Precision Devices – New Bedford, MA*

- Supported the R&D-to-production transition of high-voltage capacitor dielectric fluids, contributing to process scale-up and manufacturing readiness.
- Optimized manufacturing layout and process flow using Lean principles and statistical process control, reducing production costs by 15%.
- Authored SOPs and work instructions for material handling and process scale-up to support technician training and manufacturing consistency.
- Investigated material- and process-related failures through root cause analysis, deviation support, and CAPA implementation to improve process reliability.
- Modeled and analyzed process behavior using data-driven and statistical approaches to improve system performance and efficiency.

**R&D Technician** Mar 2023 to May 2023  
*Factorial Energy – Woburn, MA*

- Fabricated Li-metal batteries in controlled glovebox and dry-room environments to support experimental evaluation under varied operating conditions.
- Reduced material waste by 20% through redesign of the inventory system.

**Downstream Purification Engineer** Jul 2022 to Mar 2023  
*Eurofins Scientific – Framingham, MA*

- Executed purification workflows for therapeutic antibodies and enzymes using AKTA Avant systems with Unicorn software, achieving 95–99% purity.
- Applied DOE methods to optimize buffer conditions and flow rate for improved yield and purity at lab scale.
- Performed membrane filtration and chromatography studies to support pilot-scale development and manufacturing scale-up.
- Collected and interpreted analytical data using SDS-PAGE, HPLC/MS, GC/MS, and UV-Vis to confirm purity, identity, and process performance.

## EDUCATION

**M.S. Materials Science & Engineering** GPA: 3.50  
*Worcester Polytechnic Institute – Worcester, MA* 2024 to 2026

**B.S. Chemical Engineering; B.S. Applied Mathematics** GPA: 3.16  
*University of Rhode Island – Kingston, RI* 2019 to 2022

**Minors:** Chemistry, Physics

**Relevant Coursework:** Advanced Process Engineering (MATLAB/Simulink simulation)

## TECHNICAL SKILLS

**Modeling:** MATLAB, Simulink, Python, dynamic systems, numerical methods, DFT, molecular dynamics

**Tools:** ASPEN Plus Dynamics, Linux, VASP, GPAW, LAMMPS, DOE, SPC, process optimization